

2. Requirement Analysis

Functional Requirements

1. User registration & login (JWT-based authentication)
2. Add/view/delete loan records (loan amount, type, bank, interest rate, EMI, overdue months)
3. AI-generated debt negotiation strategy based on loan details
4. AI-generated settlement prediction (estimated discount percentage)
5. AI-generated negotiation email (ready to send to the lender)
6. Financial health summary (total debt, average overdue months, health status)
7. Dashboard with aggregated data (stats, charts)
8. AI strategy history and debt timeline view
9. User profile update

Non-Functional Requirements

- **Security:** Passwords hashed with bcrypt; JWT used for session management
- **Performance:** API responses under 5 seconds (including AI calls)
- **Usability:** Simple, clean UI accessible to non-technical users
- **Scalability:** RESTful API design allows future mobile app integration

Tech Stack

Layer	Technology
Frontend	React.js + Vite
Backend	FastAPI (Python)
Database	SQLite + SQLAlchemy ORM
AI	Google Gemini API (gemini-flash-latest)
Authentication	JWT (python-jose) + bcrypt (passlib)
Deployment	Render (backend), Vercel/Netlify (frontend)

User Requirements

- Users must be able to register/login securely
- Users must be able to add multiple loans and view them in a list
- Users must receive actionable, easy-to-understand AI advice (not generic)
- Users must be able to generate a ready-to-send negotiation email